

```
1 package designpatternsandprinciple.ques7;  
2  
3 public interface Stock {  
4  
5     void registerObserver(Observer observer);  
6  
7     void removeObserver(Observer observer);  
8  
9     void notifyObservers();  
10 }  
11
```

```
package designpatternsandprinciple.ques7;  
  
public interface Observer {  
    void update(String stockName, double price);  
}
```

```
3 import java.util.ArrayList;
4 import java.util.List;
5
6 public class StockMarket implements Stock {
7
8     private List<Observer> observers = new ArrayList<>();
9
10    private String stockName;
11    private double price;
12
13    @Override
14    public void registerObserver(Observer observer) {
15        observers.add(observer);
16    }
17
18    @Override
19    public void removeObserver(Observer observer) {
20        observers.remove(observer);
21    }
22
23    @Override
24    public void notifyObservers() {
25
26        for (Observer observer : observers) {
27            observer.update(stockName, price);
28        }
29    }
30
31    public void setStockData(String stockName, double price) {
32        this.stockName = stockName;
33        this.price = price;
34
35        System.out.println("\nStock Updated:");
36        System.out.println(stockName + " = Rs." + price);
37
38        notifyObservers();
39    }
40 }
```

```
1 package designpatternsandprinciple.ques7;
2
3
4
5 public class MobileApp implements Observer {
6
7     private String userName;
8
9     public MobileApp(String userName) {
10         this.userName = userName;
11     }
12
13     @Override
14     public void update(String stockName, double price) {
15
16         System.out.println(
17             "Mobile App [" + userName + "] received update: "
18             + stockName + " = Rs." + price);
19     }
20 }
21
```

```
1 package designpatternsandprinciple.ques7;
2
3 public class WebApp implements Observer {
4
5     private String userName;
6
7     public WebApp(String userName) {
8         this.userName = userName;
9     }
10
11     @Override
12     public void update(String stockName, double price) {
13
14         System.out.println(
15             "Web App [" + userName + "] received update: "
16             + stockName + " = Rs." + price);
17     }
18 }
19
```

```
1 package designpatternsandprinciple.ques7;
2
3
4
5 public class ObserverTest {
6
7     public static void main(String[] args) {
8
9         StockMarket stockMarket = new StockMarket();
10
11         Observer mobileUser = new MobileApp("Ambika");
12         Observer webUser = new WebApp("User1");
13
14         stockMarket.registerObserver(mobileUser);
15         stockMarket.registerObserver(webUser);
16
17         stockMarket.setStockData("TCS", 4200);
18
19         stockMarket.setStockData("Infosys", 1800);
20     }
21 }
```

Stock Updated:

TCS = Rs.4200.0

Mobile App [Ambika] received update: TCS = Rs.4200.0

Web App [User1] received update: TCS = Rs.4200.0

Stock Updated:

Infosys = Rs.1800.0

Mobile App [Ambika] received update: Infosys = Rs.1800.0

Web App [User1] received update: Infosys = Rs.1800.0